

Abel Mendoza

Failure Analysis Technician | Electronics Test and Validation | Hardware Troubleshooting
Anaheim, CA
abelxmendoza@gmail.com | (562) 322-2299
linkedin.com/in/abelxmendoza

Professional Summary

Electronics and robotics technician with hands-on experience performing hardware validation, troubleshooting, failure isolation, root cause analysis, component-level rework, and corrective action across industrial electronics and autonomous robotic systems. Experienced with DC-DC converter testing, relay logic validation, embedded hardware bring-up, robotics deployment support, and diagnostic testing using programmable electronic loads, bench power supplies, multimeters, and network diagnostic tools. Strong background working with electromechanical systems, documenting findings, and collaborating with engineering teams to improve reliability, quality, and product performance.

Technical Skills

Failure Analysis and Troubleshooting: Root Cause Analysis, Failure Isolation, Hardware Validation, Board-Level Debugging, Reliability Analysis, Diagnostic Testing, Corrective Actions

Electronics and Embedded Systems: DC-DC Converters, Relay Logic Circuits, PIC32 Microcontrollers, Raspberry Pi, FPGA Tooling, Embedded Linux, UART, SPI, I2C, GPIO

Electronics Rework and Assembly: Component-Level PCB Rework, Through-Hole Soldering, Resistor and Capacitor Replacement, Inspection, Repair Verification

Test Equipment: Digital Multimeters, Bench Power Supplies, Programmable Electronic Loads, Optical Power Meters, Continuity Testing

Programming: Python, C++, Bash, JavaScript

Documentation: Technical Reports, Validation Records, Test Procedures, Engineering Documentation

Professional Experience

Electronics Test and Validation Technician

Apr 2026 – May 2026

RLH Industries, Orange, CA

- Performed root cause investigation, troubleshooting, and validation testing on industrial electronic assemblies including DC-DC power converters, relay systems, and fiber optic communication hardware
- Performed component-level PCB rework, replacing and re-soldering resistors, capacitors, and other electronic components that failed quality inspection or electrical testing, improving product reliability and test yield
- Conducted board-level failure isolation using multimeters, programmable electronic loads, continuity testing, embedded diagnostics, and known-good reference hardware
- Executed DC-DC converter validation including voltage regulation testing, load characterization, tolerance verification, and thermal drift analysis
- Validated optical communication systems operating at 850nm and 1310nm wavelengths using calibrated optical power meters
- Diagnosed hardware anomalies, documented findings, and communicated results to engineering teams to support corrective actions and product reliability improvements
- Performed firmware flashing, embedded hardware bring-up, and industrial Ethernet device deployment using PIC32-based systems, MPLAB ICD3, MPFS2, USB Blaster, and Quartus

Robotics Deployment Technician

Jan 2025 – Dec 2025

Armstrong Robotics, Hollywood, CA

- Diagnosed and resolved hardware, sensor, communication, and control system failures during live deployment of autonomous robotic systems
- Performed failure investigation and corrective actions across mechanical, electrical, and electromechanical subsystems operating in production environments
- Replaced, calibrated, and validated robotic grippers and automation hardware to maintain operational reliability
- Collaborated directly with engineering teams to identify recurring failure modes and improve system performance

- Documented operational issues and troubleshooting procedures to support maintenance and future deployments

Field Robotics Agent

Mar 2025 – May 2025

Serve Robotics, Los Angeles, CA

- Investigated hardware, sensor, navigation, and connectivity failures affecting autonomous delivery robots in live field operations
- Performed systematic troubleshooting and verification testing to restore operational status
- Escalated technical findings to engineering teams and contributed data supporting reliability improvements

Technical Projects

Omega-1 Autonomous Robotics Platform

Raspberry Pi 4B | Python | Embedded Linux

- Integrated motor controllers, sensors, power systems, and embedded hardware into a custom autonomous robotics platform
- Diagnosed communication, timing, and signal integrity issues affecting servo control, sensor performance, and system stability
- Performed hardware validation and debugging across power distribution, motor control, ultrasonic sensing, and vision subsystems
- Developed diagnostic tools to monitor system health and validate hardware performance

Education

California State University, Fullerton

May 2024

Bachelor of Science in Computer Science

Rio Hondo College

Electrical and Electronics Coursework

Additional

U.S. Citizen - ITAR/EAR Eligible

FAA Part 107 Certified

Languages: English (Native), Spanish (Conversational)

Comfortable working in laboratory, manufacturing, production, and field environments